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The Skincare Investigation · 03

What the Ingredient List Reveals

The chemistry of exclusion: an ingredient deep-dive

Strip away the marketing and read the formulation: what decades of research and the products themselves reveal about skincare for melanin-rich skin.

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FIRST PUBLISHED 4 SEPTEMBER 2026 · THIS EDITION NOVEMBER 2026

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THE CHEMISTRY OF EXCLUSION. AN INGREDIENT DEEP-DIVE



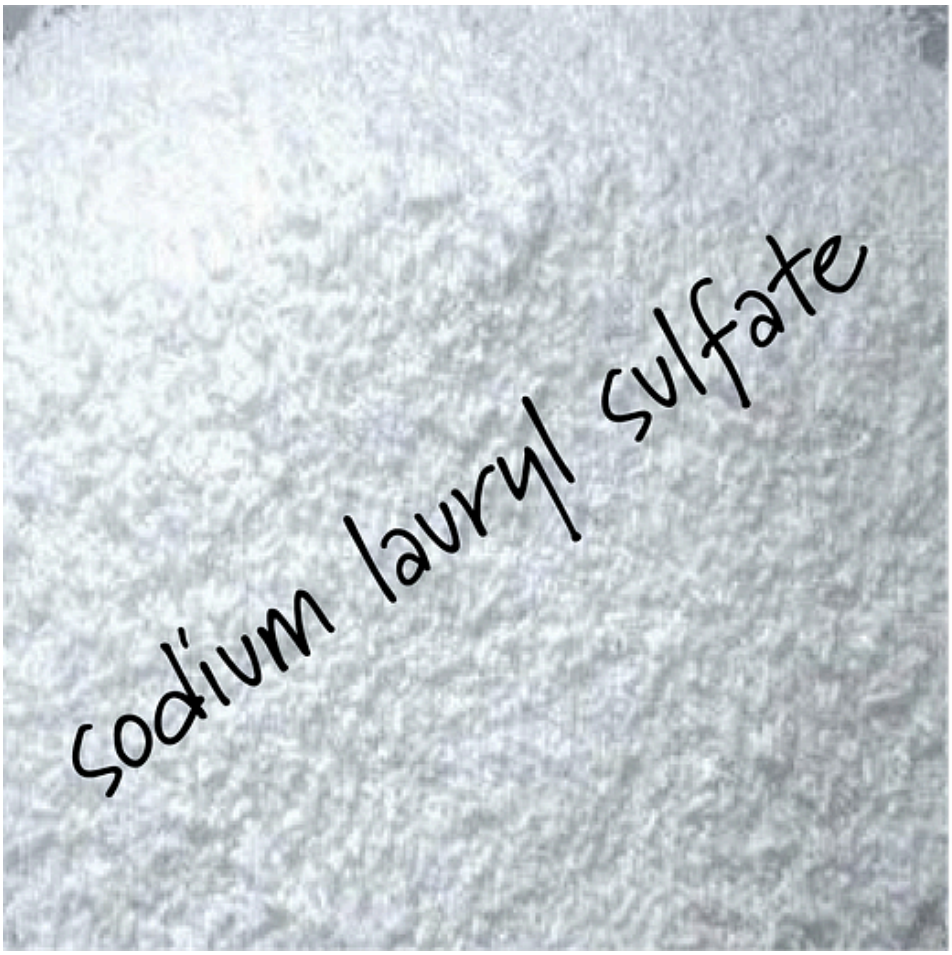
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SEP 04, 2026

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THE CHEMISTRY OF EXCLUSION. AN INGREDIENT DEEP-DIVE

The Ingredient List No One Reads

In 1988, two dermatological researchers. Enzo Berardesca at the University of Pavia and Howard Maibach at the University of California, San Francisco conducted an experiment that should have changed the skincare industry. They applied ***sodium lauryl sulfate,*** the foaming agent found in virtually every cleanser, shampoo, and body wash on the market, to the backs of nineteen volunteers. Nine were white. Ten were Black. All were men.[51]



The design was more sophisticated than a simple patch test. Each volunteer was tested at three sites: skin left untreated, skin pre-treated with occlusion, and skin that had been stripped of its natural lipids beforehand. A laboratory approximation of what a harsh cleanser does to the skin barrier every time it is used. Responses were measured by laser Doppler velocimetry, transepidermal water loss, and stratum corneum water content.

What they found was published in the journal Contact Dermatitis under the unassuming title ***"Racial differences in sodium lauryl sulphate induced cutaneous irritation: black and white."*** The results were unambiguous: Black and white skin did not respond to the same irritant

in the same way. Most tellingly, when the skin had been stripped of its lipids before exposure, the measured responses levelled out in white subjects, but not in Black subjects, whose skin continued to register pronounced reactions across the measurement conditions. Strip the barrier first, and the difference between the two groups became more visible, not less. Which is precisely the condition in which most skincare products meet Black skin.

> **This was 1988. Thirty-eight years ago.**

To understand why prestige skincare fails Black consumers, you must understand what the ingredients actually do. Not what the marketing claims. Not what the apothecary-style packaging implies. What the chemistry accomplishes when it meets melanin-rich skin.

The Surfactant Problem

Sodium Lauryl Sulfate is what chemists call a surfactant. A compound that lowers the surface tension between substances, allowing oil and water to mix. It’s the reason your cleanser foams. The reason your shampoo lathers. The reason your body wash feels like it’s doing something



It's also, as dermatologists put it, "too good at its job."

SLS doesn't merely remove dirt. It strips natural oils from the skin's surface. The sebum that functions as the skin's first line of defence against moisture loss. For consumers whose skin produces abundant

sebum, this stripping effect is temporary. The skin recovers. The oils return.

Here the beauty press usually reaches for a claim that melanin-rich skin produces less sebum and loses water faster, and is therefore more fragile at baseline. It is worth being exact, because that claim does not survive contact with the literature.

On sebum the measured differences run the other way: studies of sebum excretion and sebaceous gland size have repeatedly found equal or greater output in subjects of African descent, and the largest cross-ethnic comparison recorded the highest casual sebum levels of seven groups in African American subjects..[48]

On barrier function the evidence is genuinely contradictory. A 2003 review counted six of eight studies finding higher transepidermal water loss in Black skin, other work found the reverse or no difference at all, and a 2021 review concluded the studies were small, methodologically inconsistent, and of unestablished clinical relevance.[49]

The argument does not require that claim, and it is stronger without it. What ***Berardesca and Maibach*** measured in 1988 was not a fragile baseline. It was a response to an irritant and the divergence between the two groups appeared not on intact skin but on skin whose barrier had already been stripped.

That is the finding that matters, because of what irritation does next. On melanin-rich skin, inflammation activates melanocytes, and activated melanocytes deposit pigment. Post-inflammatory hyperpigmentation follows acne in roughly 65% of Black patients against 25% of white ones. The same insult, on two skins, produces two outcomes: on one, a week of redness; on the other, a dark patch that can take months to clear and may not clear evenly.[16]

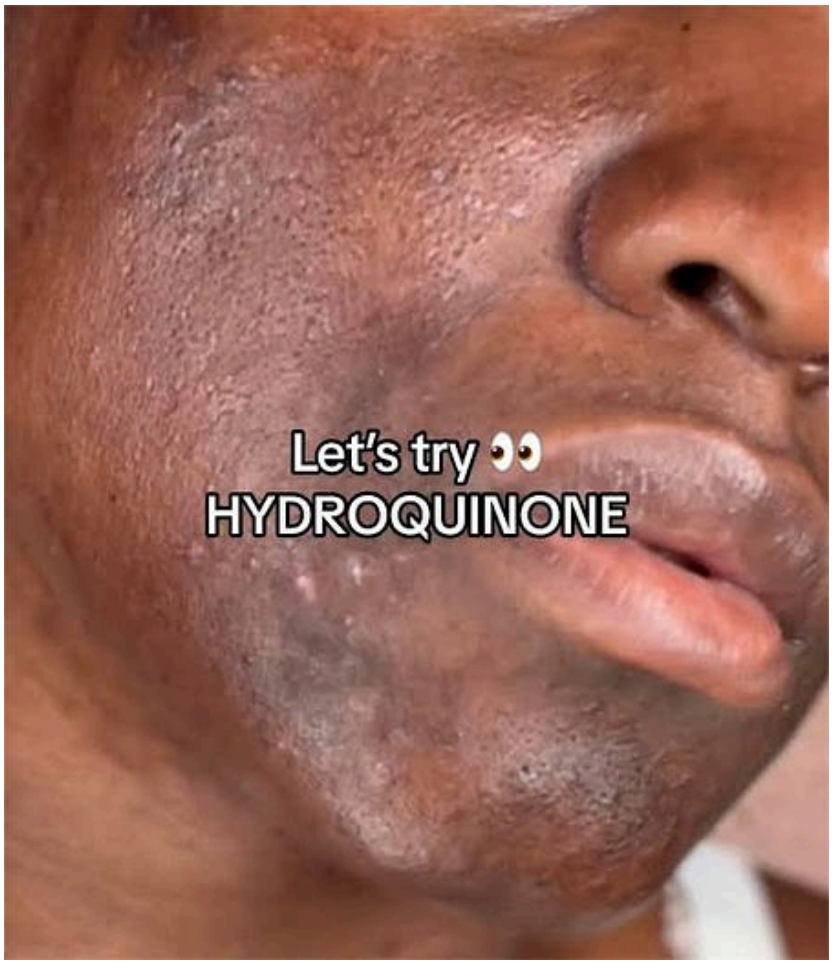
So a wash built around the most reliable irritant in cosmetic science does not need a more fragile barrier to do disproportionate harm. It only needs to irritate. The consequence is asymmetric even when the insult is identical and that asymmetry, not a sebum differential, is what the industry has never priced in.

The Hydroquinone Paradox

If sodium lauryl sulfate represents negligent formulation, ***hydroquinone*** represents something darker: an ingredient that harms the consumers it purports to help.

Hydroquinone is the gold standard treatment for ***hyperpigmentation.*** The darkened patches that form when melanocytes overproduce pigment in response to inflammation, injury, or sun exposure. Dermatologists have prescribed it for sixty years. The FDA approved it for over-the-counter use until 2020, when concerns forced a reclassification. The European Union has banned it entirely in cosmetic products.

The concern has a name: exogenous ochronosis. It is a paradoxical darkening of the skin. Blue-black patches, described in clinical literature as “caviar-like,” that develop at the sites where hydroquinone was applied. The condition is disfiguring. It is difficult to treat. And it is, in many cases, permanent.



A 2021 systematic review in the International Journal of Dermatology analysed 126 patients who developed ochronosis from hydroquinone use. The findings should have prompted industry-wide recalibration:[52]

> **Ochronosis was most often reported in people of African descent: 45.2%.**
>
> **Black races comprised 55.5% of cases.**
>
> **Fitzpatrick skin types V–VI accounted for 52.4% of cases.**

Let that register. The skin-lightening ingredient most commonly prescribed to Black consumers causes permanent disfigurement most commonly in Black consumers. The population most likely to seek treatment for hyperpigmentation is the population most likely to be harmed by the standard treatment.

The chemistry explains why.

Tyrosinase the enzyme that produces melanin appears to interact with hydroquinone in ways that researchers are still working to understand. A 2025 study in the British Journal of Dermatology settled part of the mechanism: exogenous ochronosis does not work the way the endogenous form does, and tyrosinase activity is not merely implicated but essential to its development after long-term hydroquinone use.[46]

The more active your melanocytes, the higher your risk. The more melanin your skin produces, the more likely the “cure” becomes worse than the disease.

This is not a fringe concern. This is the most commonly prescribed treatment for a condition that affects the majority of Black patients who visit dermatologists. And the industry’s response has been to continue selling it. In concentrations up to 8% when compounded. While placing the burden of monitoring for rare side effects on consumers who were never told how un-rare those effects actually are.

The Retinoid Instruction Gap

Retinoids: Vitamin A derivatives including tretinoin, retinol and adapalene are the closest thing dermatology has to a universal ingredient. They accelerate cell turnover, stimulate collagen, treat acne, and fade hyperpigmentation.



And, contrary to what a good deal of beauty writing assumes, they work perfectly well on melanin-rich skin. The clinical literature is clear on this: topical retinoids are generally well tolerated in pigmented skin, with little evidence that darker-skinned patients face an increased risk of irritation. They are a first-line treatment for exactly the conditions that disproportionately affect Black patients melasma, post-inflammatory hyperpigmentation, pseudofolliculitis barbae, keloids.[45]

> The problem is not the molecule. It is what happens on the occasions when irritation does occur.

Retinoid dermatitis. The redness, peeling and dryness of the initial adjustment period is common to all skin. On melanin-rich skin it carries a consequence it does not carry elsewhere: irritation triggers melanocytes, and triggered melanocytes leave marks. The same paper that finds retinoids well tolerated notes that “retinoid dermatitis can

induce post-inflammatory hyperpigmentation," and that "attempts should be made to reduce its occurrence by modifying treatment regimens in patients with pigmented skins."[45]

The modification is not complicated. Start at a lower concentration. Apply less frequently at first. Choose a cream or lotion over a gel. Introduce it gradually and moisturise alongside it. A dermatologist treating diverse patients does this as a matter of course, and has done since at least 2011, when that guidance was published.

Now read the box on a prestige retinol serum.

It says apply nightly. It says retinol 1%. It does not say that if irritation appears you should reduce frequency rather than push through it, and it does not say that the cost of pushing through is different depending on how much melanin your skin produces. The clinical caution exists. It is thirteen years old. It simply does not travel from the journal to the packaging.

> That gap is the failure. Not a dangerous ingredient sold to unsuspecting consumers an effective ingredient sold without the one instruction that would let a substantial share of its users get the benefit without the marks. The information costs nothing to print. Nobody prints it.

The Disclosure Gap

If the retinoid problem is an instruction that never reaches the box, the chemical peel market shows what it looks like when someone bothers to print it.

Glycolic acid, the most common alpha hydroxy acid used in chemical peels, accelerates exfoliation by dissolving the bonds between dead

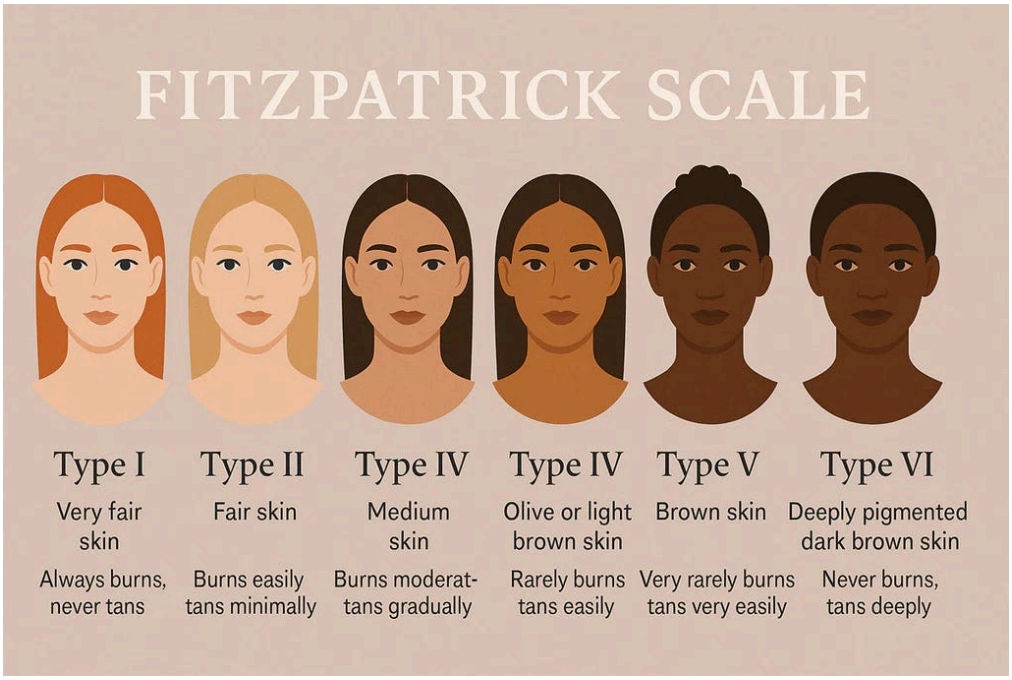
skin cells. At low concentrations 5% to 10% it brightens dull complexions and fades surface pigmentation. At professional concentrations 50% to 70% it can resurface skin, treat acne scars, and address melasma.



It can also cause burns, scarring, and permanent hyperpigmentation. And the risk increases dramatically with skin tone.

"Deep chemical peels cannot be used in dark-skinned patients owing to the risk of prolonged hyperpigmentation," states a clinical review in the Journal of Cutaneous and Aesthetic Surgery. "Even medium-depth peels need to be used with extreme caution."[53]

The Fitzpatrick scale, which classifies skin types from I (always burns, never tans) to VI (never burns, tans profusely) exists precisely to help clinicians calibrate treatments for diverse patients. Types IV, V, and VI require different concentrations, different timing, different post-treatment protocols. The peer-reviewed literature is unambiguous: the same peel that rejuvenates Type II skin can permanently disfigure Type V skin.



> Which raises an obvious question about what consumers are told. The answer is not what you would expect.

MedPeel sells professional-strength acid peels direct to the public. Its glycolic 70% listing states that due to the increased risk of hyperpigmentation, users with darker skin tones Fitzpatrick IV, V and VI, named by number are advised to take extra precaution and more thorough skin prep. Another listing for the same strength goes further and says it is not recommended for darker skin at all. Its Jessner kit sets

a hard limit: for medium to dark skin tones, do not apply beyond four to six layers, and prep the skin for a week beforehand.[47]



That is not perfect guidance. But it is Fitzpatrick-specific, it is actionable, and in one case it declines the sale.

Now consider who is not doing this. A budget supplier shipping acid in a plastic bottle names the skin types at risk and tells you when to stop. The prestige house charging forty pounds for a retinol serum, sold in the same category to the same consumers, prints nothing. The shower gel whose second ingredient is a known irritant prints nothing. ***The hotel amenity in the marble bathroom prints nothing.***

The information is not expensive. It is not proprietary. It is not difficult to phrase. A company with a fraction of the margin has already worked out

how to say it. What the prestige market lacks is not the knowledge or the means, but any reason to think the omission will cost it anything.

The Cumulative Failure

No single ingredient is the villain. ***Sodium lauryl sulfate*** has legitimate applications. ***Hydroquinone,*** properly monitored, can treat hyperpigmentation safely. Retinoids remain valuable tools. Glycolic acid delivers real results. Even alcohol has its place in formulation science.

The failure is systemic. It lies in formulating products as though all skin behaves identically. In ignoring research that documents racial differences in skin physiology. In marketing active ingredients without the guidance necessary for diverse consumers to use them safely. In treating "universal" as a formulation philosophy rather than a marketing fiction.

The 1988 study on sodium lauryl sulfate should have been the beginning of a conversation. Instead, it was filed away. Cited occasionally in academic literature, ignored entirely by the industry it should have transformed. Thirty-eight years later, Black consumers are still purchasing cleansers that strip their skin, treatments that trigger hyperpigmentation, and "luxury" products that leave them drier than before application.

> The ingredients are listed on the bottles. The research is published in the journals. The warnings are documented in the clinical literature.

No one is reading them. At least, no one who formulates the products.

The Price of Ignorance

Consider the economic calculation embedded in current formulation practices.

A prestige brand can develop a single cleanser formulation, manufacture it at scale, and market it globally as “suitable for all skin types.” The economies are obvious: one SKU, one production line, one supply chain. Testing across Fitzpatrick types would add cost. Formulating variants for different skin physiologies would add complexity.

The business case for exclusion is straightforward.

But the costs are not borne by the brand. They are externalised to the consumers whose skin the products damage.

The Black woman whose daily wash disrupts her barrier function will likely purchase additional products to address the resulting dryness. Moisturisers, serums, barrier repair treatments. She becomes a more valuable customer, not despite the initial product’s failure, but because of it.

The industry profits from the problem it created.

The Black man who uses a retinol serum without guidance appropriate to his skin tone may develop post-inflammatory hyperpigmentation. He will seek treatments for that hyperpigmentation. Perhaps hydroquinone, with its attendant risks; perhaps expensive laser treatments; perhaps more “prestige” products that promise to fade dark spots. Each purchase compounds his initial misdirection. Each transaction enriches the industry that failed him.

This is not conspiracy. It is simply incentive alignment. The prestige skincare industry has no financial motivation to formulate correctly for

Black consumers. The consequences of formulating incorrectly. Dryness, irritation, hyperpigmentation. Generate additional revenue. The system functions exactly as designed.

Breaking that system requires recognising what is actually being sold. Not skincare. Not solutions. Not luxury. What is being sold is the appearance of inclusion. Diverse faces in campaigns, shade ranges that extend beyond beige, packaging that speaks the language of premium quality. Without the substance of inclusion in the products themselves.

> **The ingredient list tells the truth the marketing conceals. Water first. Sodium lauryl sulfate. Alcohol denat. Hydroquinone at concentrations dermatologists would caution against. Retinol without guidance for diverse skin. Glycolic acid without Fitzpatrick-specific warnings.**

The labels are there.

The formulations are disclosed.

The exclusion is hiding in plain sight.

All you have to do is read.

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04 — The Luxury Hotel Bathroom Has a Problem

The ingredient list tells us what the formulation does. Next, the investigation moves into the bathrooms of some of the world's most expensive hotels — and asks why prestige amenities are chosen without evidence that they work for every guest.

A Note on Method

The body of this article is selected from the original skincare investigation I created. The wording has been preserved; the editing lies in what has been included and where the article ends. The series label, subtitle, section packaging and “Next” bridge are editorial additions.

Product and ingredient information cited here was drawn from manufacturers’ own pages and the clinical literature recorded in the master research. Manufacturer formulations and product guidance can change; readers should compare any current label with the formulation or listing discussed here. Where a claim could not be supported, it was not carried into this article.

Source Notes

[16] Perkins et al., acne-induced post-inflammatory hyperpigmentation prevalence (65% African American, 48% Hispanic, 25% Caucasian); see also “Postinflammatory Hyperpigmentation,” StatPearls, NCBI Bookshelf.

[45] Halder RM et al. “Topical Retinoids for Pigmented Skin.” Journal of Drugs in Dermatology. 2011;10(5):483–489. PMID: 21533293. Department of Dermatology, Howard University College of Medicine.

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[47] MedPeel product listings for Glycolic Acid 70% Peel, Glycolic Acid 70% Deep Strength, Alpha Beta Deep Peel Duo and the Glowing Complexion Peel Kit. medpeel.com. Accessed July 2026.

[48] Sebum and sebaceous-gland comparisons: Rawlings AV. “Ethnic skin types: are there differences in skin structure and function?” International Journal of Cosmetic Science. 2006;28(2):79–93; Mahé YF

et al. “The worldwide diversity of scalp seborrhoea.” International Journal of Cosmetic Science. 2017 (n=1,325); and “Age and ethnic variations in sebaceous lipids.” Dermato-Endocrinology. 2013;5(2). The studies reviewed do not support lower sebum output in African-descent skin.

[49] Wesley NO, Maibach HI. “Racial (ethnic) differences in skin properties: the objective data.” American Journal of Clinical Dermatology. 2003;4(12):843–860; and “Racial/Ethnic Variations in Skin Barrier: Implications for Skin Care Recommendations in Skin of Color.” Journal of Drugs in Dermatology. 2021;20(9):932.

[51] Berardesca E, Maibach HI. “Racial differences in sodium lauryl sulphate induced cutaneous irritation: black and white.” Contact Dermatitis. 1988;18(2):65–70. DOI: 10.1111/j.1600-0536.1988.tb02741.x. PMID: 3365962.

[52] Ishack S, Lipner SR. “Exogenous ochronosis associated with hydroquinone: a systematic review.” International Journal of Dermatology. 2022;61(6):675–684. First published online 6 September 2021. DOI: 10.1111/ijd.15878. PMID: 34486734. Review of 56 articles describing 126 patients.

[53] Sarkar R et al. “Chemical Peels for Melasma in Dark-Skinned Patients.” Journal of Cutaneous and Aesthetic Surgery. 2012;5(4):247–253. PMID: 23378706. Guidance notes that deep peels are unsuitable in dark-skinned patients because of prolonged hyperpigmentation risk and that medium-depth peels require extreme caution.

ABOUT THIS RESEARCH

Diversity in Crisis: *The Skincare Investigation*

A six-part investigation into how the beauty industry's promises on inclusion compare with what it actually does, written by Boyowa Olugbo, Managing Director of Praxis Brands.

01	The $9.4 Billion Customer Beauty Keeps Undervaluing	PUBLISHED 11 AUGUST 2026
02	The Skin Skincare Forgot to Study	PUBLISHED 30 AUGUST 2026
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05	When Perfumers Play Dermatologist	PUBLISHED 20 OCTOBER 2026
06	The Shelf Space Struggle	FORTHCOMING

How it is researched

Factual claims carry numbered notes. Ingredient lists are taken from manufacturers' own published information, with the date accessed recorded, and cross-checked against at least two independent sources. A claim that rests on a single source is removed rather than qualified.

Our interests, stated

We are not a neutral party. Praxis Brands manufactures for beauty and fragrance brands and invests in them, and some brands this research examines compete with businesses we work with or may work with. This research is not a lobbying position: where any part argues for a change in policy or regulation in our name, that position is approved and recorded under our Responsible Lobbying Policy before it is published.

If you think we are wrong

Write to boyowa@praxisbrands.co.uk with what you think is wrong and, where you can, the evidence. We will check it, correct the research if we got it wrong, and record the correction at praxisbrands.co.uk/research.

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